

Huanqi Wen

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EDUCATION

- **University of Notre Dame** *South Bend, USA*
PhD student in Mathematics 08/2026 - present
- **Sun Yat-sen University** *Guangzhou, China*
M.S. in Mathematics 09/2023 - 06/2026
 - **Advisor:** Prof. Lei Song
 - **Master's Thesis:** The N_p property of line bundles on toric varieties
- **Sun Yat-sen University** *Guangzhou, China*
B.S. in Mathematics, Minor in Economics 09/2019 - 06/2023
 - **Undergraduate Thesis:** On the Minkowski sum of lattice points in lattice polytopes

RESEARCH INTERESTS

Algebraic geometry (syzygies, toric geometry, tropical geometry).

PREPRINTS

- Lei Song, **Huanqi Wen**. (2026). **On Property N_p of line bundles on smooth projective toric varieties**, arXiv preprint, [arXiv: 2606.30160](https://arxiv.org/abs/2606.30160)
- Lei Song, **Huanqi Wen**, Zhixian Zhu. (2026). **The integral closedness of lattice simplices with large lattice length**, arXiv preprint, [arXiv: 2606.16348](https://arxiv.org/abs/2606.16348)
- Lei Song, **Huanqi Wen**, Zhixian Zhu. (2024). **On Covering Simplices by Dilations in Dimensions 3 and 4**, arXiv preprint, [arXiv:2404.02495](https://arxiv.org/abs/2404.02495)

RESEARCH EXPERIENCE

- **Positivity of Line Bundles on Toric Varieties** *09/2023 - present*
Supervised by Prof. Lei Song
 - **Problem:** motivated by Mukai's conjecture, study whether the pair (X, L) of a projective toric variety X and an ample line bundle corresponding to a polytope P has the N_p properties.
 - **Topics:** HHL resolution, Bondal-Thomsen collection, derived category, Klyachko's classification of toric vector bundles, stability of syzygy bundles.
 - **Results:** established a criterion for the N_p property of line bundles on smooth projective toric varieties under additional conditions; proved that lattice simplices with lattice length $L(P) \geq n - 1$ are integrally closed.
- **Minkowski Sum of Lattice Points in Lattice Polytopes** *09/2022 - 09/2023*
Supervised by Prof. Lei Song
 - **Problem:** motivated by Oda's conjecture, study which lattice polytopes are normal, specifically whether the lattice points in an $n + 1$ dilation of a polytope P can be written as the sum of a lattice point in P and a lattice point in an n dilation of P .
 - **Results:** proved that simplices with large lattice length in dimensions 3 and 4 are normal by constructing coverings with dilated simplices; implemented computer programs (Python, MATLAB) to verify the covering property and normality of lattice polytopes.

AWARDS AND SCHOLARSHIPS

- **Graduate Scholarship** *2023, 2024, 2025 three times*
Sun Yat-sen University
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** *2022*
Organized by the Consortium for Mathematics and Its Applications (COMAP)
- **Undergraduate Scholarship** *2020, 2021, 2022 three times*
Sun Yat-sen University

ACADEMIC TALKS

- **Graded Syzygies** 2025 Spring
Syzygies Seminar, Sun Yat-sen University [\[notes\]](#)
- **A Series of Lectures on Toric Varieties and Relevant Classical Papers** 2024 Spring - 2024 Fall
Toric Geometry Seminar, Sun Yat-sen University [\[notes\]](#)
- **Introduction to Schemes** 2024 Spring - 2025 Spring
Algebraic Geometry Seminar, Sun Yat-sen University

CONFERENCES PARTICIPATION

- **2025 Summer Research Institute in Algebraic Geometry**, Colorado State University, USA 07/2025
- **Stable Reduction of Algebraic Curves (Mini-course)**, Xiamen University, China 05/2025
- **China Southeastern Algebraic Geometry Symposium VII**, Sun Yat-sen University, China 12/2024
- **2024 International Summer School of Algebraic Geometry**, Fudan University, China 07/2024

TEACHING EXPERIENCE

- **Graduate Teaching Assistant of Linear Algebra**, Sun Yat-sen University 2025 Fall

SKILLS

- Programming: C++, Python
- Mathematical tools: MATLAB, SageMath, Macaulay2